MIL-P-60852A(MU)
21 October 1970
SUPERSEDING
MIL-P-60852(MU)
24 October 1969

MILITARY SPECIFICATION

PROPELLANT, GRAIN, INHIBITED

1. SCOPE

1.1 This specification covers Propellant, Grain, Inhibited for use in conjunction with Catapult, Aircraft Ejection Seat M8, M9, M10, XM12, M19, XM25, XM37, and XM38.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal.form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	Ammunition and Special Weapons, General Specification for

Naval Ordnance

OS11312	Double Base Sheet Propellant, Type HE-N12

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

Munitions Command

F8597594	Grain, Propellant, Inhibited, M8, XM12, and M19

FSC 1376

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

DRAWINGS (cont 'd)

F10551434	Grain, Propellant, Inhibited, Assy, XM38
F8597595	Grain, Propellant, Inhibited, M9 and XM37
F8597596	Grain, Propellant, Inhibited, M10 and XM25
C8858559	Box, Packing, Ammunition, for Grain, Propellant, Inhibited, M8, M9, M10, XM12, XM25, M19, XM37 or XM38
C8858560	Carton, Packing, Ammunition, for Grain, Propellant, Inhibited, M8, M9, M10, XM12, XM25, M19, XM37 or XM38
IEL8597594	Inspection Equipment List for Grain, Propellant, Inhibited, M8, XM12 and M19
IEL8597595	Inspection Equipment List for Grain, Propellant, Inhibited, M9, XM37 and XM38
IEL8597596	Inspection Equipment List for Grain, Propellant, Inhibited, M10 and XM25

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications. (See 6.1)

3.2 Inhibited propellant grain.- The inhibited propellant grain shall be produced using double-base sheet propellant, Type HE-N12 in accordance with OS11312.

3.3 Functional.

3.3.1 The grain shall not fail to fire.

3.3.2 Ballistics.- The propellant grain when fired by an acceptable cartridge or secondary Igniter in an acceptable M8, M9, M10, XM12, M19, XM25, XM37 or XM38 Aircraft Ejection Seat Catapult (or an approved facimile thereof) shall be capable of producing the performances and requirements specified in Tables I and II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

3.3.2 (cont'd)

TABLE I

Temperature	Pressure Max (psi)	Action Time millisecond Max (see 6.3)	Pressure Time Integral Min (see 6.3)	Rocket Ignition millisecond Max (see 6.4)
-65° to -70°F M8, XM12, M19	4600	400	550 psi/sec	50
-65° to -70°F M9 and XM37	4600	475	550 psi/sec	50
-65° to -70°F M10, XM25	4600	450	550 psi/sec	50
65° to 75°F M8, XM12, M19	4600	350	550 psi/sec	12
65° to 75°F M9, XM37	4600	400	550 psi/sec	12
65° to 75°F M10, XM25	4600	400	550 psi/sec	12
160° to 165°F M8, XM12, M19	4600	300	550 psi/sec	12
160° to 165°F M9, XM37	4600	350	550 psi/sec	12
160° to 165°F M10 and XM25	4600	350	550 psi/sec	12

TABLE II

Temperature	Axial Thrust (Maximum)	Action Time Milliseconds, Max	Rocket Ignition Delay Milliseconds, Max
-65° to -70°F XM38	7000	550	25
65° to 70°F XM38	7000	450	25
160° to 165°F XM38	7000	450	25

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

3.4 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings and referenced specifications.

3.4.1 Grain defects.- The inhibited grain assembly shall be free from fissures, voids, cracks, cuts, dents, scratches and grooves.

3.4.2 Foreign matter.- The propellant grain shall be free from dirt, grease, oil, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspections shall be in accordance with MIL-A-2550.

4.2 First Article Sample.

4.2.1 Initial Production Sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of twelve propellant grains, inhibited. The sample shall be manufactured using the same materials, equipment processes and procedures as used in regular production.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

4.3.1.1 Submission of product.- The product shall be presented in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- A lot shall consist of an identifiable group of propellant grains consisting of:

a. Component produced by a homogeneous process.

b. Materials from not more than one lot.

c. End inhibitors from not more than one lot.

d. Inhibiting tape from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All non-conforming material shall be rejected. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1-300	100% examination	-	-	-	-
301-800	50	0	1	1	2
801-1800	90	1	2	2	3
1801-10000	200	2	3	3	4

4.3.2.1 Propellant Grain, Inhibited-Dwgs. F8597594, F8597595, F8597596, or F10551434 as applicable:

Defect	Method of Inspection
CRITICAL	
1. Grain or inhibitor defects (cracks, grooves cuts or scratches.)	Visual
2. Head end or end inhibitor missing or improperly bonded.	Visual
3. Inhibitor tape missing, incomplete or improperly applied or bonded.	Visual
4. Internal inhibitor missing or incomplete.	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

4.3.2.1 (cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of end inhibitor with inside diameter.	Gage
102. Perpendicularity of ends with outside diameter	Gage
103. Diameter inside, max.	Gage
104. Diameter inside, min.	Gage
105. Diameter of head end counterbore	Gage
106. Diameter of small counterbore	Gage
107. Depth of head end counterbore	Gage
108. Depth of small head end counterbore	Gage
109. Excessive ovality of outside diameter	Gage
110. Variation in wall thickness.	Gage
111. Foreign matter	Visual
MINOR	
201. Diameter outside	Gage
202. Total length	Gage
203. Radius missing or incorrect	Visual

4.3.2.2 Carton, Packing, for Grain, Propellant Inhibited M8, M9, M10, XM12, XM25, M19, XM37 or XM38 Dwg. C8858560:

CRITICAL
None defined.

MAJOR
None defined.

MINOR	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.3 Box, Packing, for Grain, Propellant, Inhibited, M8, M9, M10, XM12, M19, XM37 or XM38 Dwg. C8858559:

CRITICAL
None defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

4.3.2.3 (cont'd)

Defect	Method of Inspection
MAJOR	
None defined.	
MINOR	
201. Contents loose.	Manual/Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Number of cartons incorrect	Visual
207. Marking missing, incorrect or illegible	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Functional.- The following test shall be performed by the Government.

4.3.3.1.1 Ballistics.- The twelve propellant grains, inhibited, of the initial production sample (or a subsequent sample of 12 propellant grains, inhibited, whenever a change has been made in the process or the material or the inhibitor used in regular production) and 6 propellant grains inhibited from each subsequent production lot shall be temperature conditioned as specified in Table III and 4.4.1.1.1 and then tested in accordance with 4.4.1.1. Failure of any propellant grain, inhibited to comply with the requirements of 3.3 shall be cause for rejection of the lot.

TABLE III

Temperature	Sample Size	
	Initial Production Sample	Production Lot
-65° to -70°	4	2
65° to 75°	4	2
160° to 165°	4	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

4.3.4 Inspection Equipment.- The examination and tests shall be made using equipment listed on the appropriate IEL.

4.4 Test methods and procedures.

4.4.1 Functional tests.

4.4.1.1 Ballistics.- The sample shall be installed in an appropriate catapult or approved simulated equal. The head of the catapult shall be statically supported and the test weight shall be attached to the mounting lugs of the launcher and allowed to move freely. The sample shall be ballistically tested within 10 minutes after removal from the conditioning chamber. When this time is exceeded, the sample unit shall be reconditioned in accordance with 4.4.1.1.1.

4.4.1.1.1 Temperature conditioning.- A standard M8, M9, M10, XM12, M19, XM25, XM37 or XM38 catapult loaded with a dummy grain and containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy grain to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.- Packing and marking shall be in accordance with Drawings C8858559 and C8858560.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification, and the applicable propellant grain drawing number. (See 3.1)

6.2 Action time is defined as the time between the points where the pressure reaches 500 psi at the start of burning and the point where the pressure falls below 1000 psi at the end of burning on the pressure time record.

6.3 Pressure time integral is defined as the area under the pressure time record bounded by the action time limit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

6.4 Rocket ignition delay is defined as the time between the sudden drop on the acceleration time record of the booster section and the point at which the pressure time record of the rocket section reaches 500 psi.

6.5 Supercession data.- This specification includes the requirements specification MIL-P-60852(MU) dated 24 October 1969.

Custodian

Army - MU

Preparing Activity

Army - MU

Project No. 1376-A083

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R256

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

DIRECT GOVERNMENT CONTRACT SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

5. SUBMITTED BY (Printed or typed name and activity - Optional)

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
Department of the Army

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-T1000
Philadelphia, Pa. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:01Z
Check the source to verify that this is the current version before use.